

Measuring Progress on Long-Tailed Dental Radiograph Segmentation: Leakage, Ceilings, and Noise Floors

Abstract

We report a systematic negative result on a 31-class panoramic dental radiograph corpus: across nineteen training configurations spanning boundary-aware objectives, class-balanced losses, frequency-aware DETR training, an architectural change to the mask head, and decoupled classifier retraining, no configuration improves on a stock baseline beyond seed-level variation.

The contribution is not the null itself but the apparatus required to establish it, each element of which caught a confident wrong answer that would otherwise have been reported. (1) The corpus cannot be evaluated as distributed: 95.9 % of its test partition shares source images with training. (2) Perceptual hashing alone is not a leakage test on this modality, with a false-positive rate near 100 %. (3) The mask representation imposes a measurable ceiling that is *not* the binding constraint; a higher-resolution head with 878,432 *fewer* parameters performs 0.72 pp worse, as predicted. (4) Distance metrics conditioned on per-model subsets invert their own sign once compared on the intersection of scored cases. (5) Group means over low-support classes fabricate results: in one case 98 % of a +0.53 pp mAP gain came from a single class with *two* test instances. (6) The published logit-adjustment constant does not transfer from $\sim 100:1$ to 34,320:1 imbalance.

1 Introduction

Long-tailed instance segmentation and detection on clinical imagery is an active area, and the standard toolkit—class-balanced losses, logit adjustment, repeat-factor sampling, boundary-aware objectives—is well established on natural-image benchmarks. We set out to apply that toolkit to a 31-class dental radiograph corpus with 34,320:1 imbalance and measure the contribution of each component.

None of it worked. That is worth reporting, because the reasons are specific, measurable, and mostly not about the methods. The corpus as distributed cannot support evaluation at all. The architecture imposes a ceiling that is real but not binding. The metrics the field uses for contour quality invert their sign under a denominator that differs per model. And group means over classes with single-digit support will manufacture a headline on demand—twice, in our case, before we caught them.

We regard contributions (2), (4) and (5) as the most transferable: each is a measurement protocol that any group working on long-tailed clinical segmentation can apply, and each caught an error in our own work that would have survived review.

2 Related Work

Long-tailed recognition. Class-balanced loss weighting via effective number of samples [2] and logit adjustment [10] are the standard remedies. Kang et al. [5] showed representation and classifier learning should be *decoupled*, with rebalancing applied at the classifier stage; Wang et al. [14] adapt this to instance segmentation. Repeat-factor sampling [4] rebalances at the data level, and Seesaw loss [13] rescales gradients per class pair.

Our results bear on the *regime* these were validated in. Menon et al. [10] tune $\tau = 1.0$ on CIFAR-LT and ImageNet-LT at roughly 100:1 to 1000:1 imbalance. At 34,320:1 the same constant applies a +11.47 logit shift and destroys the classifier. We find the failure monotonic in τ , which localises the problem to magnitude rather than principle.

High-quality mask prediction. PointRend [9] recasts segmentation as rendering, refining adaptively at uncertain points. RefineMask [15] fuses fine-grained features in stages. Mask Transfuser [7] represents masks as a quadtree and corrects only error-prone nodes. All three exist because coarse mask prediction is a known failure mode. We measure that failure mode directly rather than assuming it, and find it is *not* what limits our model—a result invisible without the model-free ceiling measurement of Section 4.

Boundary-aware objectives. Boundary loss [8] integrates the prediction against a precomputed distance map; HD loss [6] estimates Hausdorff distance directly; Boundary IoU [1] is an evaluation metric restricted to a contour band. Our band-Dice objective is a synthesis of known components—morphological gradient as a differentiable edge extractor, soft Dice [11] as the overlap objective—and we say so rather than claiming a new loss family.

Augmentation. Copy-paste [3] reliably helps on natural images. We find it harmful here, and propose a mechanism: in panoramic radiographs anatomical position is part of the class definition, so a finding pasted into an anatomically impossible location teaches the model to detect the paste rather than the pathology.

3 The Corpus Cannot Be Evaluated as Distributed

Two further defects: a spurious `croen` category at id 12 in the test COCO file shifts every higher class id out of alignment

image files	13,932
distinct source images	6,395
test files whose source appears in train/valid	1,516 / 1,580 (95.9 %)
genuinely unseen test sources	64
patients in both train and test	194

Table 1: The distributed split. Augmentation was applied before partitioning rather than after, so any metric computed on it principally measures memorisation.

check	impl. A	impl. B
exact duplicate groups (SHA-256)	0	0
NCC-confirmed near-duplicates	0	0
dHash candidates examined	441,240	792,811
rejected as look-alikes	all	all
scope	test vs rest	all pairs

Table 2: Disjointness verified twice by independent implementations. Maximum NCC across 792,811 candidate pairs is 0.9334, against a 0.98 confirmation threshold.

with train and valid, and `data.yaml` points `test:` at the validation images.

Replacement. Patient-grouped and class-stratified: 9752 / 2090 / 2090, with test coverage rising from 13 of 31 classes to 29.

3.1 Perceptual Hashing Alone Is Not a Leakage Test

On natural images a dHash Hamming distance below ~ 5 bits is strong evidence of duplication. On panoramic radiographs it is nearly meaningless: every image is the same anatomy, same framing, same grayscale statistics, so unrelated studies routinely fall within 5 bits.

Run without pixel confirmation, dHash reports **792,811 leaking pairs on a split we independently verify as clean**—a false-positive rate near 100 %. Any audit of this modality that stops at perceptual hashing will condemn a clean split, or worse, be tuned until it stops complaining.

Two-stage detection—dHash as a cheap candidate generator, normalised cross-correlation as confirmation—separates the cases cleanly. Genuine re-encoded duplicates score $NCC \geq 0.99$; distinct studies stay below 0.94.

3.2 Statistical Power

Sixteen of 31 classes appear in fewer than 10 validation images; two appear in none; eleven tail classes score exactly zero. **Image counts, not instance counts, are the effective sample size.** No long-tail claim is supportable on this corpus, and we make none.

prototype grid	mean Dice	cannot reach IoU 0.75
160×160 (stock)	0.8963	17.7 %
320×320	0.9492	5.4 %
640×640	0.9901	0.5 %

Table 3: Model-free ceiling over 20,601 instances. Root canal treatment is capped at IoU 0.622 and caries at 0.721—both *below* the 0.75 threshold, excluding 4,493 instances from AP75 before training begins.

4 The Representation Ceiling, and Why It Is Not the Constraint

YOLOv8-seg predicts 32 coefficients per instance and re-constructs masks as a linear combination of prototypes at input/4— 160×160 at `imgsz = 640`. The median instance here is **6 px on that grid**; 68 % are under 8 px.

We bound what any model could achieve by round-tripping every ground-truth mask through the grid and scoring it against itself.

The model’s behaviour agrees: mask AP50 reaches 87 % of box AP50, mask AP75 only 49 %.

4.1 The Inference We Initially Drew Was Wrong

This looks like a diagnosis. It is not.

ceiling at the current grid	0.8963
achieved by the model	0.6969
headroom already available	0.1994
additional headroom from doubling	0.0529

Table 4: The model sits at 78 % of the ceiling it already has.

We tested it. A prototype head fed from P2 (stride 4) rather than P3 (stride 8) yields 320×320 prototypes from genuine high-resolution features, using sub-pixel convolution [12] to keep the wide tensor off the high-resolution grid, with **878,432 fewer parameters** than stock so any gain could not be a capacity artifact. Against a matched control at identical budget and protocol: **−0.72 pp mAP**, **−0.49 pp AP75**.

The ceiling is real and worth reporting. It is not the binding constraint, and the check that establishes this—comparing achieved performance against the ceiling—costs one line of arithmetic. We ran it after the experiment. It should be run before.

5 Two Metric Failures That Produce Confident Wrong Answers

5.1 Distance Metrics on Per-Model Subsets

HD95 and ASSD are undefined when either mask is empty, so implementations average them over cases where both are non-

metric	ref.	bnd.	difference	sep.
Dice	0.6953	0.6969	+0.0017	no
IoU	0.5685	0.5706	+0.0021	no
boundary F	0.7857	0.7881	+0.0024	no
HD95	66.36	68.66	+2.30 (worse)	yes
ASSD	18.40	18.82	+0.43 (worse)	no

Table 5: Recomputed on the 5,385 cases both models scored, paired, with the per-case difference bootstrapped over images. Correcting the denominator did not attenuate the effect—it *inverted the sign*.

empty. **That denominator differs per model.** A model that misses a hard structure drops that case from its own average and can post better distances by predicting less.

Averaged per model, our boundary objective appeared **17.3 % better on HD95 and 24.6 % better on ASSD, with disjoint confidence intervals**—the strongest result in the project.

Protocol. Any metric conditioned on a per-model subset must be compared on the intersection of cases both models scored, paired, with coverage reported alongside. Reporting distances without coverage permits a model to win by predicting less.

5.2 Group Means over Low-Support Classes

Two headline gains, in two independent projects, dissolved under decomposition. For detection, decoupled classifier retraining gave +0.53 pp mAP and +1.25 pp tail—positive on every metric, with the largest gain exactly where a long-tail method should deliver.

observed mAP delta	+0.53 pp
largest contributor	one class, 2 test instances
its contribution	+0.52 pp
share of total	98 %
excluding that class	+0.01 pp
on 18 classes with ≥ 10 instances	6 improve, 12 worsen

Table 6: Decomposition of the detection gain.

Protocol. Report per-class deltas with support counts alongside any group mean, and state the result restricted to classes with adequate support. A group AP over classes with single-digit support is not a measurement.

6 Results

The τ sweep gives -6.05 , -2.85 , -0.24 pp at $\tau = 1.0, 0.5, 0.25$: monotonic recovery, never reaching baseline. The catastrophe is a mis-scaled constant; the deficit is not.

model	mAP	AP50	AP75	tail
baseline	0.1051	0.2590	0.0687	0.0365
boundary alone	0.1071	0.2585	0.0680	0.0441
weighting + boundary	0.1007	0.2430	0.0650	0.0261
weighting alone	0.0973	0.2369	0.0612	0.0252

Table 7: Segmentation, held-out test. The nominal +0.21 pp of the boundary arm decomposes to three classes with 4, 8 and 8 test instances.

arm	mAP	tail
baseline	0.1625	0.0368
denoising only	0.1633	0.0455
plain reweighting	0.1461	0.0270
unified ($\tau = 1.0$)	0.1020	0.0000

Table 8: Detection, validation, matched budget. The unified treatment loses 4.41 pp to plain reweighting, which loses 1.64 pp to changing nothing. Tail AP is exactly zero in all unified arms.

Divergence is patterned. Every arm applying the adjustment inconsistently diverged with `inf` or `NaN` cost matrices in the Hungarian matcher; every consistent arm was stable. Consistency prevents divergence without producing accuracy.

7 The Mask Deficit Is Coefficient Prediction, Not Representation

The 20-point gap between what the model achieves (Dice 0.6969) and what its prototype grid allows (0.8963) had no attribution. Prototype-based heads reconstruct each mask as $\sigma(\mathbf{c}^\top P)$, so either the learned 32-dimensional basis cannot represent these shapes, or the basis is adequate and the co-efficient head fails to locate the right point in it. Those call for opposite fixes.

We separate them without training. For every ground-truth instance we solve, in closed form, for the coefficient vector that best reconstructs it from the model’s own prototypes,

$$\mathbf{c}^* = \arg \min_{\mathbf{c}} \|P^\top \mathbf{c} - y\|^2, \quad (1)$$

over the instance’s box crop, where $y \in \{-1, +1\}$ is the ground-truth mask. Thresholding $P^\top \mathbf{c}^*$ gives the best mask any coefficient head could produce from that basis.

quantity	Dice
grid ceiling (what the resolution allows)	0.8963
oracle coefficients, full resolution	0.8659
oracle coefficients, at prototype resolution	0.9636
what the trained model achieves	0.6969

This closes the loop on every negative result above. Boundary terms and comparator losses reshape the *objective*; higher prototype resolution enlarges the *representation*. Neither is the binding constraint, which is why neither moved the number,

source of the 0.1994 gap	Dice lost	share
prototype basis cannot represent it	0.0304	15 %
coefficient head fails to find it	0.1690	85 %

Table 9: 4123 instances. The basis is not the constraint: handed optimal coefficients the model jumps from 0.70 to 0.87.

and why the P2-fed head performed worse rather than better—it added resolution to a basis that was never the limit.

The head capacities make this concrete. In YOLOv8x-seg the coefficient branch bottlenecks 320 to 80 channels and carries 1.33 M parameters, against 2.26 M for the prototype generator and 7.41 M for classification. The smallest of the three heads is responsible for 85 % of the mask deficit.

8 Noise Floor (Measured)

The reference configuration was trained at three seeds, everything else identical. Determinism is exact on this pipeline, so seed choice is provably the only source of variation.

seed	mAP	AP50	AP75	head	tail
42	0.1055	0.2549	0.0661	0.2815	0.0101
1337	0.1056	0.2573	0.0679	0.2837	0.0117
2024	0.1037	0.2568	0.0629	0.2823	0.0105
sd	0.0010	0.0013	0.0025	0.0011	0.0008

Table 10: Noise floor at 2 sd: mAP ± 0.21 , AP50 ± 0.26 , AP75 ± 0.50 , head ± 0.22 , tail ± 0.17 pp.

Three consequences, and they are not the same verdict. The seven-arm ablation spans a 0.36 pp range, *inside* the band—selecting a winner on mAP was selecting noise. The best segmentation arm on test is $+0.21$ pp, sitting *exactly at* the threshold. But the isolated boundary effect on AP75 is $+0.70$ pp against a ± 0.50 pp floor, so it **exceeds** noise on validation at 50 epochs, and then reverses on test at 100 epochs. That is a **transfer failure, not a noise result**: the objective does what its mechanism predicts under the conditions it was measured in, and that does not survive a change of split and schedule.

9 Discussion

The standard long-tail and boundary-aware toolkit does not transfer to this corpus, and most of the reasons are not about the methods. Two are about the data: it cannot be evaluated as shipped, and 16 of 31 classes lack the support to measure anything. Two are about measurement: distance metrics and group means both produce confident wrong answers under conditions easy to satisfy accidentally. One is about regime: a published constant validated at 100:1 does not survive 34,320:1.

Only one is about a method—the band-Dice objective produces the AP75/AP50 signature its mechanism predicts, and

that signal does not survive to test.

We think the protocols of Sections 3.1, 5.1 and 5.2 are the durable output. Each is cheap, each applies to any long-tailed clinical segmentation study, and each caught an error in our own work that would otherwise have been published.

10 Limitations

Single seed on all reported arms; the three-seed noise floor is measured above. Single corpus, single backbone family per task. No comparison against prior published results, because none exist for this dataset. No external benchmark, and therefore no long-tail claim in the title, abstract or conclusions. Two detection cells are recorded as diverged rather than scored.

References

- [1] Bowen Cheng, Ross Girshick, Piotr Dollár, Alexander C. Berg, and Alexander Kirillov. Boundary IoU: Improving object-centric image segmentation evaluation. In *CVPR*, 2021.
- [2] Yin Cui, Menglin Jia, Tsung-Yi Lin, Yang Song, and Serge Belongie. Class-balanced loss based on effective number of samples. In *CVPR*, 2019.
- [3] Golnaz Ghiasi, Yin Cui, Aravind Srinivas, Rui Qian, Tsung-Yi Lin, Ekin D. Cubuk, Quoc V. Le, and Barret Zoph. Simple copy-paste is a strong data augmentation method for instance segmentation. In *CVPR*, 2021.
- [4] Agrim Gupta, Piotr Dollár, and Ross Girshick. LVIS: A dataset for large vocabulary instance segmentation. In *CVPR*, 2019.
- [5] Bingyi Kang, Saining Xie, Marcus Rohrbach, Zhicheng Yan, Albert Gordo, Jiashi Feng, and Yannis Kalantidis. Decoupling representation and classifier for long-tailed recognition. In *ICLR*, 2020.
- [6] Davood Karimi and Septimiu E. Salcudean. Reducing the Hausdorff distance in medical image segmentation with convolutional neural networks. *IEEE Transactions on Medical Imaging*, 2019.
- [7] Lei Ke, Martin Danelljan, Xia Li, Yu-Wing Tai, Chi-Keung Tang, and Fisher Yu. Mask transfiner for high-quality instance segmentation. In *CVPR*, 2022.
- [8] Hoel Kervadec, Jihene Bouchtiba, Christian Desrosiers, Eric Granger, Jose Dolz, and Ismail Ben Ayed. Boundary loss for highly unbalanced segmentation. In *MIDL*, 2018.
- [9] Alexander Kirillov, Yuxin Wu, Kaiming He, and Ross Girshick. PointRender: Image segmentation as rendering. In *CVPR*, 2020.
- [10] Aditya Krishna Menon, Sadeep Jayasumana, Ankit Singh Rawat, Himanshu Jain, Andreas Veit, and Sanjiv Kumar. Long-tail learning via logit adjustment. In *ICLR*, 2021.
- [11] Fausto Milletari, Nassir Navab, and Seyed-Ahmad Ahmadi. V-Net: Fully convolutional neural networks for volumetric medical image segmentation. In *3DV*, 2016.
- [12] Wenzhe Shi, Jose Caballero, Ferenc Huszár, Johannes Totz, Andrew P. Aitken, Rob Bishop, Daniel Rueckert, and Zehan Wang. Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network. In *CVPR*, 2016.

- [13] Jiaqi Wang, Wenwei Zhang, Yuhang Zang, Yuhang Cao, Jiangmiao Pang, Tao Gong, Kai Chen, Ziwei Liu, Chen Change Loy, and Dahua Lin. Seesaw loss for long-tailed instance segmentation. In *CVPR*, 2021.
- [14] Tao Wang, Yu Li, Bingyi Kang, Junnan Li, Jun Hao Liew, Sheng Tang, Steven Hoi, and Jiashi Feng. Classification calibration for long-tail instance segmentation. *arXiv preprint arXiv:1910.13081*, 2019.
- [15] Gang Zhang, Xin Lu, Jingru Tan, Jianmin Li, Zhaoxiang Zhang, Quanquan Li, and Xiaolin Hu. RefineMask: Towards high-quality instance segmentation with fine-grained features. In *CVPR*, 2021.